



REDROB · DATA & AI CHALLENGE

# HireFit Ranker V6

## Universal strength. Battle-proof release.

A deterministic AI recruiter that reads careers and behavior — then publishes an exact, validated top-100 shortlist.

**100K**

candidates scored

**136s**

100K · 2 CPU · release

**#1**

of 673 · seven-evaluator mean

**0**

honeypots in top-100

*\* Public-field rank is a transparent local comparison, not an official leaderboard. See slide 14.*

HIREFIT RANKER

## THE PROBLEM

# Keyword filters miss the right people — and rank the wrong ones first.

The dataset is engineered to punish keyword matching. Four trap types sink a naive ranker.



### Keyword stuffers

Wrong-role profiles padded with AI skills — a Marketing Manager listing 9 AI keywords.



### Wrong-role profiles

Perfect skill lists attached to titles that never shipped a ranking system.



### Honeypots

Subtly impossible profiles — “expert” in 10 skills with 0 months used. Forced to tier 0.



### Unavailable-but-perfect

Great on paper, but 5% recruiter response and not logged in for months.



**The real task:** rank candidates the way a great recruiter would — reading career history and behavior, not counting keywords. A candidate who *built* a recommender at a product company outranks one who merely *lists* “RAG” and “Pinecone.”

# The JD lists keywords; the role is hiring for evidence.

Our whole design follows from reading the gap between what the JD says and what it means.



## What the JD says

*“5–9 years, embeddings, retrieval, ranking, NLP, deep learning, PyTorch, TensorFlow...”*

embeddings

retrieval

ranking

NLP

5–9 yrs

**The trap:** skill-keyword overlap is high for stuffers and low for the engineers who actually shipped the systems.



## What the role means



**Shipped work** — Built & deployed ranking / search / recsys in production.



**Career trajectory** — Consistent growth in relevant, product-company roles.



**Right seniority** — anchored on the JD's 5–9 band — juniors demoted hard; senior depth read from trajectory, not years arithmetic.



**Actually available** — Responsive, open to work, reachable notice period.

## MEASURED DISCIPLINE

# We attacked quality, configuration, and failure modes before shipping.

Every claim below is tied to committed output, a test, or a reproduced run.

**30 / 30**

COMPOSITES BEAT MAIN

**10K**

CORRUPT OUTPUTS  
REJECTED

**9,750**

INVALID CONFIGS REJECTED

**262**

TESTS PASSED



### Quality evidence

**Main comparison** V6 wins all 30 composites across 15 evaluator families and two missing-label policies

**Public field** #1/673 on mean7; #1/100 on strongest union; #3/322 on four-axis balance

**Exact release** one pinned input, one pinned model, one expected output hash



### Battle-proof before shipping

Release mode rejects experimental flags, validates every row and integrity count, then publishes atomically. A forced 3 GiB OOM preserved the previous output and left zero mounted temporary files.



**Honest framing: public ranks are local comparisons; official challenge weights and hidden labels are not published.**

# Deterministic at rank time — exact output guarded at release.

Six CPU-only stages, no network or GPU; V6 adds hash checks, full validation, and atomic publish.



**Final score = base score × behavioral × honeypot × disqualifier**

Guardrails are multiplicative — a strong-on-paper profile that is impossible, off-target, or unreachable is pushed down no matter how good its keywords look.

RUNS UNDER:

✓ Offline

✓ CPU-only

✓ No GPU

✓ Deterministic

✓ bm25s backend

# 33 features separate keyword lists from recruiter-plausible fit.

Each signal is computed deterministically from profile facts — and every rank decomposes back into them.



## Skills & depth

10 features

core / nice-to-have match, skill depth, endorsement trust, assessment scores, GitHub, semantic (HyRE) match



## Career evidence

11 features

product-company ratio, IR/ranking experience, production evidence, seniority, trajectory, scale, role-family depth (backend, data/BI)



## Experience fit

4 features

years-of-experience band, education, ML/AI tenure, open-source signal



## Behavior

6 features

availability, engagement, responsiveness, interview reliability, profile quality, notice



## Logistics

2 features

preferred-location match, willingness to relocate



## Multiplicative guardrails

Behavioral, honeypot & disqualifier multipliers gate the base score so traps can't buy their way to the top.

# The shortlist is real: every rank is traceable.

Live output from the released 100K pool, plus the exact feature values and guardrails behind the #1 candidate.

Ranked output

submission.csv top 6

| Rank | Candidate evidence                                                                       | YOE | Score |
|------|------------------------------------------------------------------------------------------|-----|-------|
| 1    | <b>Senior ML Engineer - Zomato</b><br>hybrid retrieval + ranking at 50M+ queries         | 7.2 | 1.000 |
| 2    | <b>Senior NLP Engineer - Niramai</b><br>semantic search + embeddings + 89% response      | 7.8 | 0.984 |
| 3    | <b>Senior AI Engineer - Apple</b><br>FAISS + fine-tuning + 30-day notice                 | 5.9 | 0.960 |
| 4    | <b>Senior ML Engineer - Flipkart</b><br>semantic search at 35M+ item scale               | 8.1 | 0.957 |
| 5    | <b>Lead AI Engineer - Razorpay</b><br>IR + vector search + learning to rank              | 6.7 | 0.932 |
| 6    | <b>Senior NLP Engineer - Salesforce</b><br>Pinecone + Elasticsearch; Amazon search depth | 8.9 | 0.931 |

Readout: all six combine production retrieval/ranking evidence, seniority, and recruiter-ready behavior.

Why #1 wins

CAND\_0018499

Senior ML Engineer at Zomato | 7.2 yrs | Noida | 15-day notice

core\_skill\_match

0.982

production\_evidence

1.000

ir\_ranking\_experience

1.000

product\_company\_ratio

1.000

career\_trajectory\_score

1.000

behavior x1.10

honeypot x1.00

DQ x1.00

**Grounded reason:** shipped retrieval/ranking; skills include Learning to Rank + Embeddings; response 0.61.

# Stuffers and honeypots never reach the shortlist.

Measured on the released pool — the guardrails do real work, not decoration.



Keyword stuffers buried

4,800+

Wrong-role profiles across the pool's 12 non-target strata — zero reach the top-100.



Honeypots blocked

53 → 0

Impossible profiles detected, then zero emitted in the top 100. Challenge DQ threshold is >10%.



Unavailable down-weighted

×0.42

Behavioral multiplier on an 8-yr profile with 12% response & not-open-to-work — pushed off the shortlist.



**Plain-language fit is captured too:** skill matching runs over the full profile text, not just the skills array — so an engineer who *describes* building a recommender (without the buzzwords) still scores on core fit.



# V6 is #1 on two broad composites — and #3 on balance.

1,367 repos discovered · 1,279 eligible · 672 valid outputs · one evaluator harness.

**#1 / 673**

SEVEN-EVALUATOR MEAN

**#1 / 100**

STRONGEST-UNION MEAN

**#3 / 322**

FOUR-AXIS BALANCE

**30 / 30**

COMPOSITES VS MAIN



## What the comparison proves

**Mean7** 0.906553 — broad agreement across seven evaluator families

**Mean15** 0.910406 — strongest available label union

**Balance** 0.876596 — strength across four distinct quality axes



## No public four-axis dominator

Every measured public competitor gives up at least one axis. V6 is not #1 on every specialist judge, but it has the strongest broad portfolio and beats main on every tested composite.



Transparent local comparison, not an official Hack2skill leaderboard. Repository availability and evaluator coverage vary.

# Universal strength beats one-metric overfitting.

The deck separates measured wins from uncertainty judges should know.

## Defensible decisions

shipped path

- **No LLM per candidate**  
100K hosted calls break the CPU/no-network budget and make reproduction fragile.
- **Dense embeddings tested, then kept off**  
NDCG@10 was flat while runtime and complexity rose, so the simpler ranker won.
- **Guardrails are multiplicative**  
Impossible, off-target, or unreachable profiles cannot buy rank with keywords.
- **Deterministic reasoning**  
Same input gives the same CSV; every sentence points to real profile facts.

## Known uncertainty

why no fake  
certainty

- **Hidden official labels**  
Official weights and hidden scores are unseen, so public-field results are evidence — not a trophy claim.
- **Specialist rankings**  
H2 #14, reviewer #115, blind estimate #20. Broad composites recover the strongest balance.
- **Modern challengers lost**  
Dense, learned-LR, and LambdaMART variants failed their declared adoption gates.
- **Deterministic fallback**  
If multiprocessing is constrained, --workers 1 preserves the exact same output.

# One release command reproduces the shortlist — and fails closed.

Built for the challenge sandbox: pinned, offline, atomic, and exact-output verified.

```
>_ python rank.py --release --workers 2 --candidates candidates.jsonl --out  
submission.csv
```

## CONSTRAINT

## CHALLENGE LIMIT

## MEASURED

Runtime

≤ 5 min (300s)

136.0 s



Memory

≤ 16 GB RAM

4.13 GiB



Compute

CPU only

CPU only



Network

off during ranking

fully offline



## Submission package



GitHub repo — tested, documented, reproducible



submission.csv — exact top-100 + grounded reasons



Challenge positioning — weights and caveats disclosed



Deck + PDF — same V6 values as the README



262 passed · 6 environment skips

# A drop-in scoring layer for Redrob — fast, auditable, and outcome-ready.

It converts job intent, career evidence, and activity signals into a recruiter-readable shortlist.

**~3 ms**

AUDIT API, MEASURED

**136 s**

FULL 100K RELEASE RUN

**0**

LLM CALLS PER RANKED  
CANDIDATE

**33**

EXPLAINABLE FEATURES



## Integration path

**JD compiler** nuanced role text becomes a deterministic scoring program

**Audit payload** features, multipliers, flags, and grounded career evidence

**Signals** response, recency, notice, reliability, and open-to-work integrate natively



## Where it grows

Indic-script normalization is localized. With real outcome labels, learned reranking can reuse the 33-feature audit surface. The durable asset is the evidence model plus the release contract.



**LLMs evaluate offline; deterministic evidence ranks where hiring decisions must be defended.**

## THE TAKEAWAY

# Battle-proof by design. Strong by evidence.



Beats main on every tested composite — 30 / 30



Exact 100K release in 136.0 s — zero traps emitted



Projected #1 · honest range #1–#3 · not an official  
leaderboard



Public-field #1 on mean7 and strongest union



262 tests; corrupt configs and OOM fail safely

## MISSION SCORECARD

# The challenge asks for an AI recruiter. V6 maps to every mission pillar.

Mission-derived weights are transparent and explicitly not official.

93.7

MISSION SCORE / 100

#1

PROJECTED PUBLIC-FIELD  
POSITION

#1—#3

HONEST UNCERTAINTY RANGE

0

OFFICIAL WEIGHTS INVENTED



### Why the score is high

Ranking quality 32.6 / 35 — broad field leadership

JD context + signals 27.5 / 30 — 33 career and behavior features

Speed + robustness 14.7 / 15 — 136 s, exact, fail-closed



### Submission complete

Code, README, methodology, architecture, ranked output, battle-proof audit, demo script, deck, and PDF now tell one V6 story.



**Final position: projected #1 against the measured public field; the official hidden leaderboard remains unknown.**